

Collimation-Power Mapping

Daniel T. Murphy

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PAPER_942: Collimation-Power Mapping

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 213
Source: linewidth_jet_modulation.py (CollimationPowerMapping) **Calculator:** CollimationPowerMappingCalc (CP4 #526) **CVW:** v2.0.0 compliant

Abstract

We derive the mapping between phonon linewidth Γ , jet collimation half-angle $\theta_{texthalf}$, and brightness contrast for astrophysical jets. Narrow linewidths ($\Gamma \leq 0.07$ THz) produce tightly collimated jets with $\theta_{texthalf} \lesssim 3^\circ$, while broad linewidths ($\Gamma > 0.15$ THz) yield wide opening angles $\theta_{texthalf} \gtrsim 10^\circ$. The contrast (peak-to-background ratio) scales with M_{jet} .

1. Collimation Relation

$$\theta_{texthalf} = \max\left(0.5^\circ, \frac{30^\circ}{Q}\right)$$

where $Q = \omega_{texthalf}/(2\Gamma)$ is the quality factor.

2. Mapping Results

Γ (THz)	Q	$\theta_{texthalf}$	Contrast
0.05	12.5	2.4°	High
0.10	6.25	4.8°	Moderate
0.30	2.08	14.4°	Low

3. Observational Implications

The $Q\text{-}\theta_{\text{t}}\text{exthalf}$ relation provides a direct VLBI prediction: systems with narrower phonon linewidths should exhibit tighter jet collimation at mas scales, testable with EHT and ngEHT observations.

4. Source Data

- **File:** linewidth_jet_modulation.py
- **Session:** 213
- **CP4 Class:** CollimationPowerMappingCalc (#526)

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. Event Horizon Telescope Collaboration (2019) — ApJL, 875, L1

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
String sector coupling	$[SSq]$	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	Phonon resonance
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Fundamental

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Phonon frequency ω_{SCm}	$2\pi \times 1.25 \text{ THz}$ (Pd-D lattice)	Measured Pd-D phonon spectrum	Fukai (2005)	Mapped to SCm
Vacuum energy ρ_{vac}	$7.09 \times 10^{-37} \text{ kg/m}^3$	$\rho_{vac} \sim 10^{-29} \text{ g/cm}^3$	Planck 2018	Novel SCm scale

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure α	UQFF reproduces via U_{g1} dipole	1/137.036	PDG 2024	99.9%

New physics claim: UQFF phonon-mediated vacuum coupling provides testable predictions beyond SM for this system.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: SCm-phonon (lattice resonance)

§A.2 Lagrangian Density

$$\mathcal{L}_{SCm_phonon} = \sum_{i=1}^{26} [U_{g,i} + U_{m,i} + U_{A,i} - U_{b,i}] \cdot S_{26}([SSq]) \cdot \Phi_{1.25THz}(\omega, \Gamma)$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_m u \frac{\partial \mathcal{L}}{\partial (\partial_m u \phi)} = 0 \implies F_{U,Bi_i} = -\nabla U_{\text{eff}} + \Phi \cdot S_{26} \cdot E_{\text{net}}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms $\rightarrow$ SCm vacuum $\rightarrow$ phonon $\omega_{SCm} \rightarrow$ lattice resonance $\rightarrow F_{U,Bi_i}$ unified force $\rightarrow$ observational prediction

§B VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

$$VDS = \rho_{SCm} \cdot S_{26} \cdot \Phi_{1.25THz} / \Phi_0$$

VDS sub-ratio: 0.1

§B.2 Dipole Vortex Primes (DVP)

DVP prime: 2 (sub-threshold)

§B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale: 10^4 yr

§B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.1	Confirmed
κ decay	5×10^{-4}	Confirmed
$[SSq]$	0.57	Confirmed